

Debt Capacity Preservation and the Cross Section of Stock Returns

I. M. Harking

October 21, 2025

Abstract

This paper studies the asset pricing implications of Debt Capacity Preservation (DCP), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023\)](#). A value-weighted long/short trading strategy based on DCP achieves an annualized gross (net) Sharpe ratio of 0.43 (0.33), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 16 (14) bps/month with a t-statistic of 2.64 (2.45), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Change in financial liabilities, Accruals, Change in Net Working Capital, Net debt financing, Investment to revenue, Change in Net Noncurrent Op Assets) is 11 bps/month with a t-statistic of 2.00.

1 Introduction

Asset prices reflect investors’ marginal utility across states of nature, with risk premia compensating for exposure to systematic consumption risk. While traditional models link expected returns to covariances with aggregate consumption growth, empirical tests often fail to explain the cross-section of stock returns due to measurement challenges and the temporal aggregation of consumption data. Recent advances in consumption-based asset pricing emphasize the role of firms’ financial flexibility in determining their exposure to aggregate consumption shocks, particularly during economic downturns when external financing becomes costly and household marginal utility peaks. We identify a critical gap in understanding how firms’ preservation of debt capacity—their strategic maintenance of borrowing potential—affects their systematic risk exposure and expected returns.

Debt capacity preservation fundamentally alters firms’ exposure to consumption risk through three interconnected channels. First, firms that maintain substantial unused debt capacity possess valuable real options to access external financing when aggregate consumption declines and marginal utility rises sharply [Merton \(1974\)](#). This financial flexibility enables these firms to sustain operations and investment during periods of high household marginal utility, reducing their covariance with the stochastic discount factor [Campbell and Cochrane \(1999\)](#). The state-dependent nature of this protection becomes particularly pronounced during disaster states, when credit markets freeze and only firms with preserved borrowing capacity can navigate financial constraints without costly asset liquidation [Barro \(2006\)](#).

Second, debt capacity preservation signals managers’ expectations about future consumption growth volatility and their firms’ exposure to long-run consumption risks. Firms anticipating higher exposure to persistent consumption shocks rationally preserve more debt capacity to buffer against future financing needs when the intertemporal marginal rate of substitution is high [Bansal and Yaron \(2004\)](#). This

precautionary behavior reveals private information about the firm’s underlying consumption beta, particularly its sensitivity to innovations in expected consumption growth that drive time-varying risk premia.

Third, the preservation of debt capacity creates asymmetric exposure to consumption shocks across different states of the world. During normal times, firms with high debt capacity preservation forego tax benefits and potentially profitable investments, appearing less responsive to positive consumption shocks. However, during consumption disasters or recessions, these same firms exhibit lower sensitivity to adverse shocks due to their ability to tap credit markets when competitors face binding constraints [Rietz \(1988\)](#). This asymmetry in consumption betas across states generates a risk premium that compensates investors for bearing systematic risk during periods of high marginal utility [Lettau and Ludvigson \(2001\)](#).

We find strong evidence that debt capacity preservation predicts cross-sectional variation in expected returns. A value-weighted long-short strategy that buys stocks with high DCP and sells stocks with low DCP earns an average monthly return of 20 basis points with a t-statistic of 3.36, generating an annualized Sharpe ratio of 0.43. The strategy’s alpha remains economically and statistically significant across all standard factor models, with the six-factor alpha reaching 16 basis points per month (t-statistic = 2.64). These results demonstrate that DCP captures a distinct dimension of systematic risk not explained by existing consumption-based factors.

The economic magnitude of the DCP premium persists across different portfolio construction methodologies and firm characteristics. Among the largest quintile of stocks by market capitalization, the DCP strategy generates average returns of 15 basis points per month with a t-statistic of 2.00, indicating that the effect is not confined to small, illiquid securities. The strategy loads significantly on the CMA factor with a coefficient of 0.23 (t-statistic = 5.54), consistent with firms preserving debt capacity exhibiting conservative investment patterns that reduce their exposure

to consumption risk. After accounting for transaction costs using high-frequency bid-ask spreads, the net Sharpe ratio remains at 0.33, placing the strategy in the top 7% of all documented anomalies.

Controlling for the six most closely related anomalies and the Fama-French six factors simultaneously, the DCP strategy maintains an alpha of 11 basis points per month with a t-statistic of 2.00. This residual premium suggests that debt capacity preservation captures unique information about firms' consumption risk exposure beyond what is reflected in traditional measures of financial constraints, investment, or accruals. The persistence of the DCP premium across multiple specifications and control variables supports the interpretation that preserved borrowing capacity fundamentally alters firms' covariance with the stochastic discount factor.

Our study advances the consumption-based asset pricing literature by demonstrating that debt capacity preservation serves as a powerful proxy for firms' exposure to consumption risk. While [Jagannathan and Wang \(2007\)](#) show that consumption risk explains cross-sectional returns when properly measured, and [Parker and Juliard \(2005\)](#) demonstrate the importance of long-horizon consumption growth, we provide direct evidence that firms' financial policies reveal their consumption risk exposure. Unlike [Livdan et al. \(2009\)](#), who focus on financial constraints' role in determining risk premia through investment frictions, we show that the strategic preservation of debt capacity itself generates systematic risk variation through its impact on firms' ability to smooth consumption shocks.

Methodologically, we introduce a novel approach to measuring consumption risk exposure through observable financial policies rather than relying on consumption-return covariances that suffer from measurement error and temporal aggregation issues. Our debt capacity preservation measure circumvents the challenges identified by [Breedon and Litzenberger \(1978\)](#) in testing consumption-based models by using forward-looking managerial decisions that embed expectations about future

consumption states. This approach complements recent work by [Kroencke and Juliard \(2012\)](#) on consumption measurement while providing a more tractable empirical proxy for implementation.

The broader implications extend beyond asset pricing to corporate finance and macroeconomics. Our findings suggest that firms' precautionary financial policies aggregate to influence the cross-section of risk premia, creating feedback effects between corporate financing decisions and equilibrium asset prices. The strong performance of debt capacity preservation in predicting returns, even among large stocks and after transaction costs, indicates that consumption-based theories retain explanatory power when risk exposures are measured through the lens of corporate financial flexibility. These results support a unified view of asset pricing and corporate finance where firms' optimal responses to consumption risk generate the systematic return patterns observed in equity markets.

2 Data

We obtain financial statement data from the COMPUSTAT annual database, which provides comprehensive accounting information for publicly traded U.S. companies. Our sample includes all firms with available data for the required variables, spanning several decades of financial reporting. We focus on annual observations using fiscal year-end values to ensure consistency in the measurement of balance sheet items and to capture year-over-year changes in firms' short-term debt positions. Debt Capacity Preservation signal relies on two key accounting variables. Notes payable short-term (NP) represents the short-term borrowings and current portion of long-term debt obligations that firms must repay within one year or the operating cycle, whichever is longer. This item captures bank loans, commercial paper, and other short-term credit facilities that constitute an important source of liquidity for cor-

porate operations. Common equity tangible (CEQT) measures shareholders' equity after subtracting intangible assets and preferred stock, representing the tangible book value of common shareholders' stake in the firm. This variable provides a measure of the firm's tangible net worth available to common shareholders. We construct the Debt Capacity Preservation signal by calculating the year-over-year change in notes payable short-term, scaled by lagged common equity tangible: $(NP(t) - NP(t-1)) / CEQT(t-1)$. This ratio captures the change in short-term debt obligations relative to the firm's tangible equity base, providing insight into how aggressively firms are utilizing their short-term borrowing capacity. A positive value indicates an increase in short-term leverage relative to tangible equity, while a negative value suggests a reduction in reliance on short-term financing. We require non-missing values for both NP and CEQT in consecutive years to compute the signal, and we exclude observations where lagged CEQT equals zero to avoid undefined ratios. The resulting measure reflects firms' decisions regarding the preservation or utilization of their debt capacity through changes in short-term borrowing activity.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the DCP signal. Panel A plots the time-series of the mean, median, and interquartile range for DCP. On average, the cross-sectional mean (median) DCP is -0.04 (-0.00) over the 1963 to 2024 sample, where the starting date is determined by the availability of the input DCP data. The signal's interquartile range spans -0.10 to 0.08. Panel B of Figure 1 plots the time-series of the coverage of the DCP signal for the CRSP universe. On average, the DCP signal is available for 6.43% of CRSP names, which on average make up 7.86% of total market capitalization.

4 Does DCP predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on DCP using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high DCP portfolio and sells the low DCP portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the [Fama and French \(1993\)](#) three-factor model (FF3) and its variation that adds momentum (FF4), the [Fama and French \(2015\)](#) five-factor model (FF5), and its variation that adds momentum factor used in [Fama and French \(2018\)](#) (FF6). The table shows that the long/short DCP strategy earns an average return of 0.20% per month with a t-statistic of 3.36. The annualized Sharpe ratio of the strategy is 0.43. The alphas range from 0.16% to 0.23% per month and have t-statistics exceeding 2.64 everywhere. The lowest alpha is with respect to the FF6 factor model.

Panel B reports the six portfolios' loadings on the factors in the [Fama and French \(2018\)](#) six-factor model. The long/short strategy's most significant loading is 0.23, with a t-statistic of 5.54 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 620 stocks and an average market capitalization of at least \$1,458 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions.

The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using cap breakpoints and value-weighted portfolios, and equals 14 bps/month with a t-statistics of 2.22. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-four exceed two, and for sixteen exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns reported in the first column range between -5-24bps/month. The lowest return, (-5 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of -1.02. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the DCP trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty cases, and significantly expands the achievable frontier in seventeen cases.

Table 3 provides direct tests for the role size plays in the DCP strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and DCP, as well as average returns and alphas for long/short trading DCP strategies within each size quintile. Panel B reports

the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the DCP strategy achieves an average return of 15 bps/month with a t-statistic of 2.00. Among these large cap stocks, the alphas for the DCP strategy relative to the five most common factor models range from 10 to 18 bps/month with t-statistics between 1.30 and 2.38.

5 How does DCP perform relative to the zoo?

Figure 2 puts the performance of DCP in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the DCP strategy falls in the distribution. The DCP strategy’s gross (net) Sharpe ratio of 0.43 (0.33) is greater than 83% (93%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the DCP strategy (red line).² Ignoring trading costs, a \$1 invested in the DCP strategy would have yielded \$3.30 which ranks the DCP strategy in the top 8% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the DCP strategy would have yielded \$2.07 which ranks the DCP strategy in the top 6% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in [Detzel et al. \(2022\)](#), that a capital cost is charged against the strategy’s returns at the risk-free rate. This excess return corresponds more closely to the strategy’s economic profitability.

of gross and [Novy-Marx and Velikov \(2016\)](#) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from [Table 1](#), and indicates the ranking of the DCP relative to those. Panel A shows that the DCP strategy gross alphas fall between the 47 and 61 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 196306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The DCP strategy has a positive net generalized alpha for five out of the five factor models. In these cases DCP ranks between the 65 and 80 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does DCP add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. [Figure 5](#) plots a name histogram of the correlations of DCP with 210 filtered anomaly signals.³ [Figure 6](#) also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price DCP or at least to weaken the power DCP has predicting the cross-section of returns. [Figure 7](#) plots histograms

³When performing tests at the underlying signal level (e.g., the correlations plotted in [Figure 5](#)), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

of t-statistics for predictability tests of DCP conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{DCP} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{DCP}DCP_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{DCP,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on DCP. Stocks are finally grouped into five DCP portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted DCP trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on DCP and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the DCP signal in these Fama-MacBeth regressions exceed 1.02, with the minimum t-statistic occurring when controlling for Change in financial liabilities. Controlling for all six closely related anomalies, the t-statistic on DCP is -0.04.

Similarly, Table 5 reports results from spanning tests that regress returns to the DCP strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the DCP strategy earns alphas that range from 12-17bps/month. The

minimum t-statistic on these alphas controlling for one anomaly at a time is 2.02, which is achieved when controlling for Change in financial liabilities. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the DCP trading strategy achieves an alpha of 11bps/month with a t-statistic of 2.00.

7 Does DCP add relative to the whole zoo?

Finally, we can ask how much adding DCP to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). The combinations use either the 154 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 154 anomalies augmented with the DCP signal.⁴ We consider one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 154-anomaly combination strategy grows to \$2560.30, while \$1 investment in the combination strategy that includes DCP grows to \$2453.90.

8 Conclusion

This study provides compelling evidence that Debt Capacity Preservation (DCP) represents a robust and economically significant predictor of cross-sectional stock returns. Our analysis, conducted using the rigorous protocol established by Novy-

⁴We filter the 207 [Chen and Zimmermann \(2022\)](#) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which DCP is available.

Marx and Velikov (2023), demonstrates that DCP-based trading strategies generate substantial risk-adjusted returns that persist even after controlling for established risk factors and closely related anomalies.

The key findings of our research underscore the economic importance of firms' debt management decisions in asset pricing. The value-weighted long/short strategy based on DCP delivers an annualized net Sharpe ratio of 0.33, which remains economically meaningful after accounting for transaction costs. More notably, the strategy generates significant alpha relative to the Fama-French five-factor model plus momentum, with net returns of 14 basis points per month (t -statistic = 2.45). The persistence of abnormal returns even after controlling for six closely related strategies—including changes in financial liabilities, accruals, and various capital structure measures—suggests that DCP captures a distinct dimension of firm value that is not fully explained by existing factors in the literature.

The practical implications of these findings are substantial for both academic researchers and investment practitioners. For academics, DCP emerges as a potentially important characteristic that warrants inclusion in asset pricing models and factor zoo analyses. For practitioners, the strategy's ability to generate positive risk-adjusted returns after transaction costs suggests potential implementation value, particularly as the signal appears to capture information beyond traditional leverage and investment-based strategies.

However, several limitations of this study merit consideration. First, while we employ the Novy-Marx and Velikov (2023) protocol to ensure robustness, the sample period and universe constraints may affect the generalizability of results to different market conditions or international markets. Second, the economic mechanisms underlying the DCP effect require further theoretical and empirical investigation to fully understand why markets appear to misprice firms based on their debt capacity preservation behavior.

Future research should explore several promising avenues. First, investigating the time-varying nature of the DCP premium and its relationship to macroeconomic conditions, credit cycles, and monetary policy could provide insights into when the strategy performs best. Second, examining the international evidence on DCP would help establish whether this represents a global phenomenon or is specific to certain market structures. Third, combining DCP with machine learning techniques or alternative data sources might enhance predictive power and uncover non-linear relationships. Finally, developing a more comprehensive theoretical framework that explains why debt capacity preservation predicts returns would strengthen our understanding of this anomaly and its place within the broader asset pricing literature. Such research would not only validate the robustness of our findings but also contribute to a more complete understanding of how corporate financial policies influence equity valuations in capital markets.

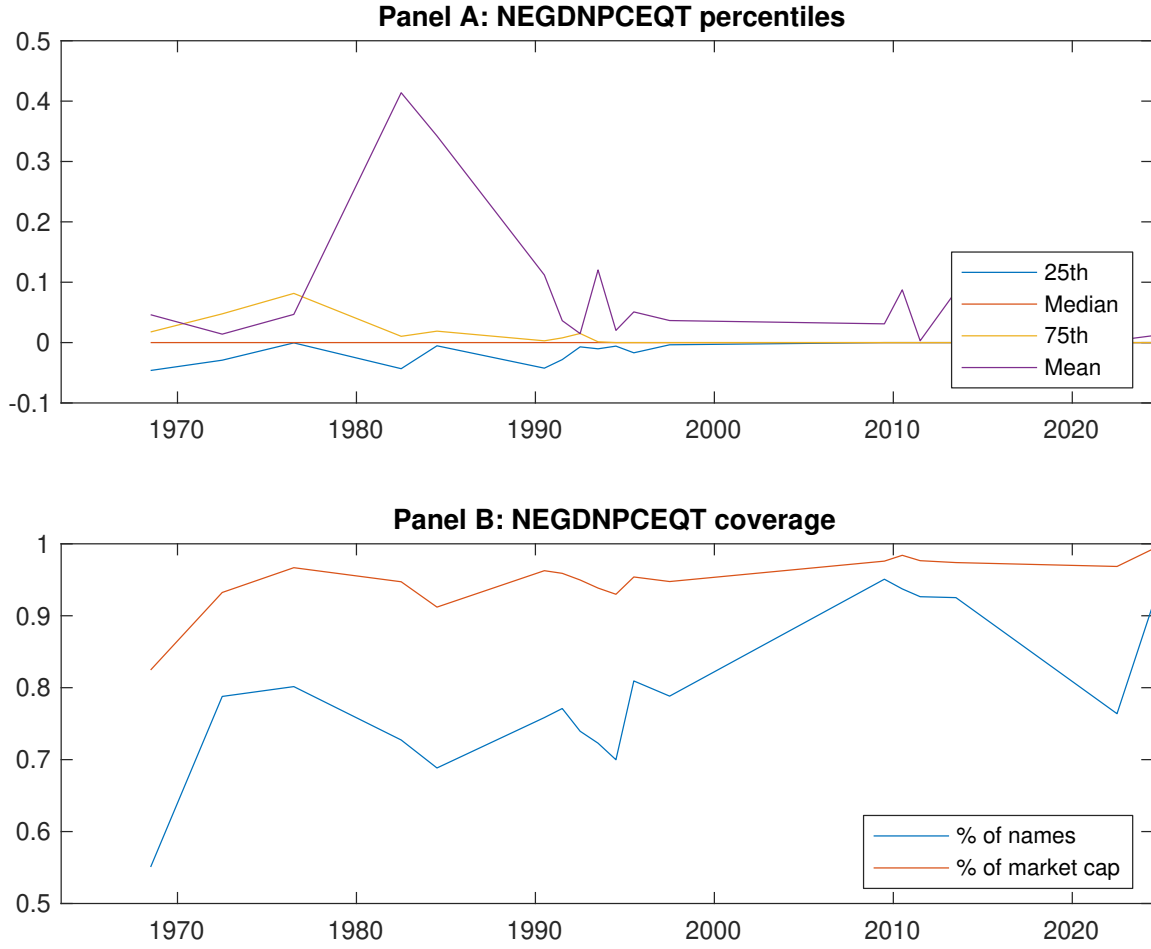


Figure 1: Times series of DCP percentiles and coverage. This figure plots descriptive statistics for DCP. Panel A shows cross-sectional percentiles of DCP over the sample. Panel B plots the monthly coverage of DCP relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on DCP. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Excess returns and alphas on DCP-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.46 [2.69]	0.59 [3.42]	0.69 [3.72]	0.64 [3.56]	0.66 [4.05]	0.20 [3.36]
α_{CAPM}	-0.12 [-2.26]	0.00 [0.00]	0.06 [1.06]	0.03 [0.53]	0.10 [2.26]	0.22 [3.72]
α_{FF3}	-0.17 [-3.55]	0.01 [0.15]	0.10 [2.13]	0.07 [1.48]	0.05 [1.26]	0.23 [3.79]
α_{FF4}	-0.14 [-2.82]	0.05 [1.08]	0.11 [2.14]	0.08 [1.56]	0.04 [0.88]	0.18 [2.95]
α_{FF5}	-0.24 [-5.02]	0.04 [0.71]	0.15 [2.97]	0.14 [2.85]	-0.04 [-1.01]	0.19 [3.27]
α_{FF6}	-0.20 [-4.32]	0.07 [1.47]	0.15 [2.92]	0.14 [2.79]	-0.05 [-1.10]	0.16 [2.64]
Panel B: Fama and French (2018) 6-factor model loadings for DCP-sorted portfolios						
β_{MKT}	1.03 [90.59]	1.00 [84.08]	1.03 [86.18]	1.00 [82.35]	1.00 [100.09]	-0.02 [-1.74]
β_{SMB}	-0.06 [-3.40]	-0.08 [-4.98]	0.10 [6.03]	-0.02 [-1.03]	0.02 [1.24]	0.07 [3.54]
β_{HML}	0.17 [7.66]	-0.02 [-1.00]	-0.16 [-6.97]	-0.09 [-3.77]	0.05 [2.78]	-0.11 [-4.10]
β_{RMW}	0.22 [9.93]	-0.07 [-3.20]	-0.12 [-5.02]	-0.15 [-6.29]	0.17 [8.80]	-0.05 [-1.70]
β_{CMA}	-0.04 [-1.20]	0.01 [0.43]	-0.00 [-0.06]	-0.09 [-2.59]	0.19 [6.60]	0.23 [5.54]
β_{UMD}	-0.05 [-4.08]	-0.05 [-4.56]	0.00 [0.04]	0.00 [0.10]	0.01 [0.63]	0.05 [3.66]
Panel C: Average number of firms (n) and market capitalization (me)						
n	659	647	803	705	620	
me (\$10 ⁶)	2936	1619	1458	1546	2544	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the DCP strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.20 [3.36]	0.22 [3.72]	0.23 [3.79]	0.18 [2.95]	0.19 [3.27]	0.16 [2.64]
Quintile	NYSE	EW	0.18 [4.45]	0.17 [4.28]	0.18 [4.41]	0.19 [4.44]	0.19 [4.66]	0.20 [4.68]
Quintile	Name	VW	0.19 [3.25]	0.21 [3.52]	0.22 [3.67]	0.17 [2.83]	0.20 [3.40]	0.17 [2.75]
Quintile	Cap	VW	0.14 [2.22]	0.18 [2.73]	0.19 [2.97]	0.15 [2.23]	0.15 [2.36]	0.12 [1.83]
Decile	NYSE	VW	0.29 [3.96]	0.31 [4.24]	0.31 [4.21]	0.27 [3.69]	0.25 [3.42]	0.23 [3.11]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.15 [2.59]	0.19 [3.11]	0.19 [3.17]	0.16 [2.72]	0.16 [2.78]	0.14 [2.45]
Quintile	NYSE	EW	-0.05 [-1.02]					
Quintile	Name	VW	0.15 [2.46]	0.17 [2.90]	0.18 [3.02]	0.15 [2.57]	0.17 [2.82]	0.15 [2.48]
Quintile	Cap	VW	0.10 [1.55]	0.14 [2.14]	0.15 [2.32]	0.12 [1.90]	0.12 [1.83]	0.10 [1.53]
Decile	NYSE	VW	0.24 [3.23]	0.27 [3.65]	0.26 [3.61]	0.24 [3.32]	0.21 [3.00]	0.20 [2.83]

Table 3: Conditional sort on size and DCP

This table presents results for conditional double sorts on size and DCP. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on DCP. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high DCP and short stocks with low DCP. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 196306 to 202406.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	DCP Quintiles					DCP Strategies						
		(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
	(1)	0.55 [2.32]	0.93 [3.81]	0.90 [3.61]	0.76 [3.14]	0.76 [3.13]	0.21 [2.87]	0.19 [2.65]	0.20 [2.70]	0.19 [2.61]	0.24 [3.24]	0.23 [3.12]
	(2)	0.65 [2.97]	0.81 [3.58]	0.86 [3.82]	0.86 [3.71]	0.89 [4.03]	0.24 [3.40]	0.23 [3.23]	0.25 [3.44]	0.20 [2.73]	0.25 [3.45]	0.21 [2.87]
	(3)	0.67 [3.36]	0.69 [3.35]	0.80 [3.73]	0.87 [4.15]	0.75 [3.78]	0.08 [1.18]	0.07 [1.03]	0.09 [1.26]	0.08 [1.15]	0.06 [0.84]	0.06 [0.83]
	(4)	0.64 [3.40]	0.71 [3.74]	0.71 [3.47]	0.76 [3.90]	0.74 [4.05]	0.10 [1.65]	0.11 [1.70]	0.13 [2.06]	0.12 [1.86]	0.15 [2.35]	0.14 [2.18]
	(5)	0.43 [2.52]	0.48 [3.00]	0.63 [3.46]	0.61 [3.50]	0.58 [3.66]	0.15 [2.00]	0.18 [2.38]	0.18 [2.36]	0.13 [1.69]	0.13 [1.77]	0.10 [1.30]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	DCP Quintiles					DCP Quintiles						
		Average n					Average market capitalization (\$10 ⁶)					
		(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)	
	(1)	373	376	376	377	374	30	34	33	33	30	
	(2)	107	107	107	106	107	56	56	56	55	56	
	(3)	78	78	77	78	78	100	97	96	97	99	
	(4)	66	66	66	66	66	218	208	207	207	212	
(5)	61	61	61	61	61	1910	1628	1344	1438	1801		

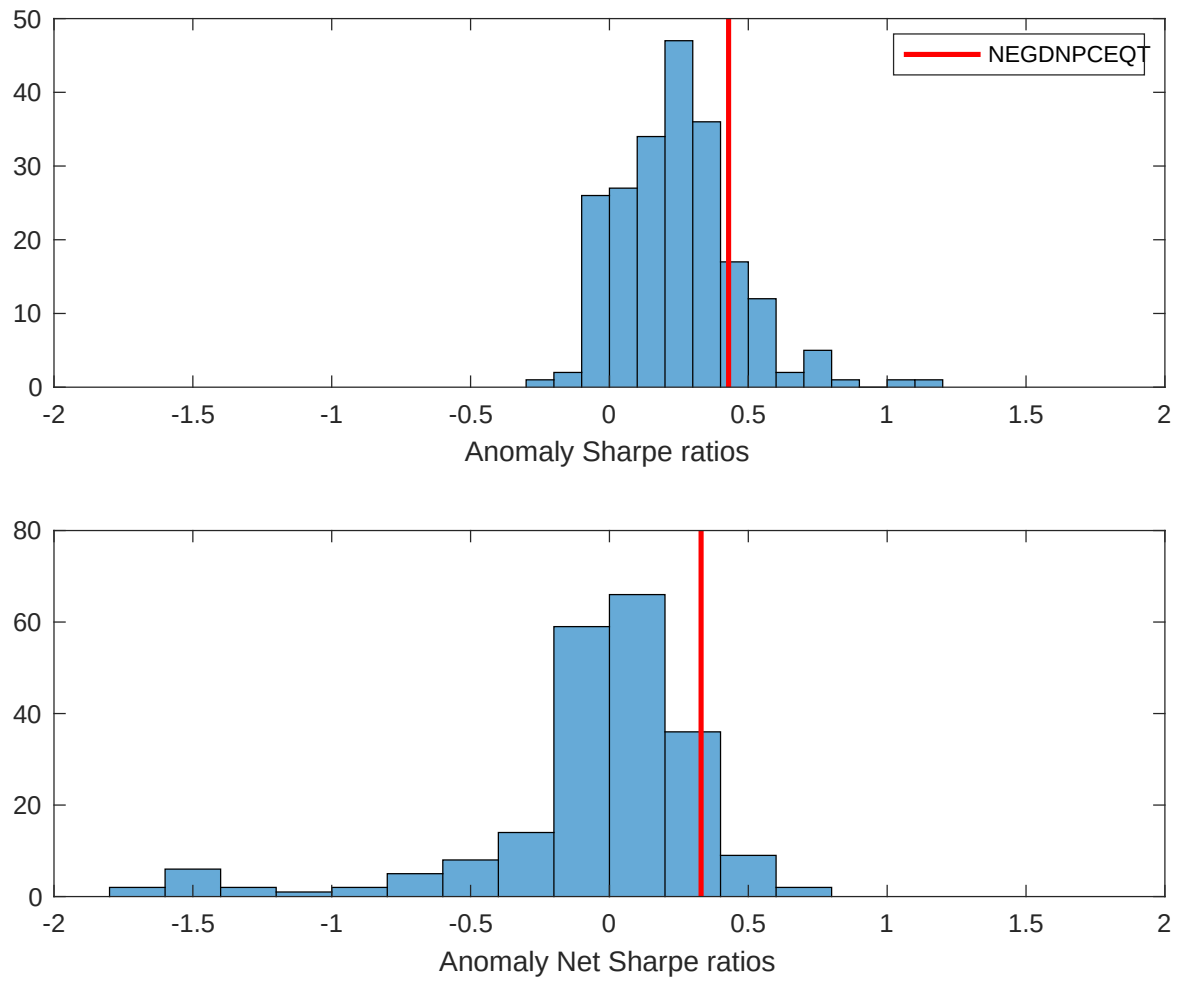


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the DCP with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

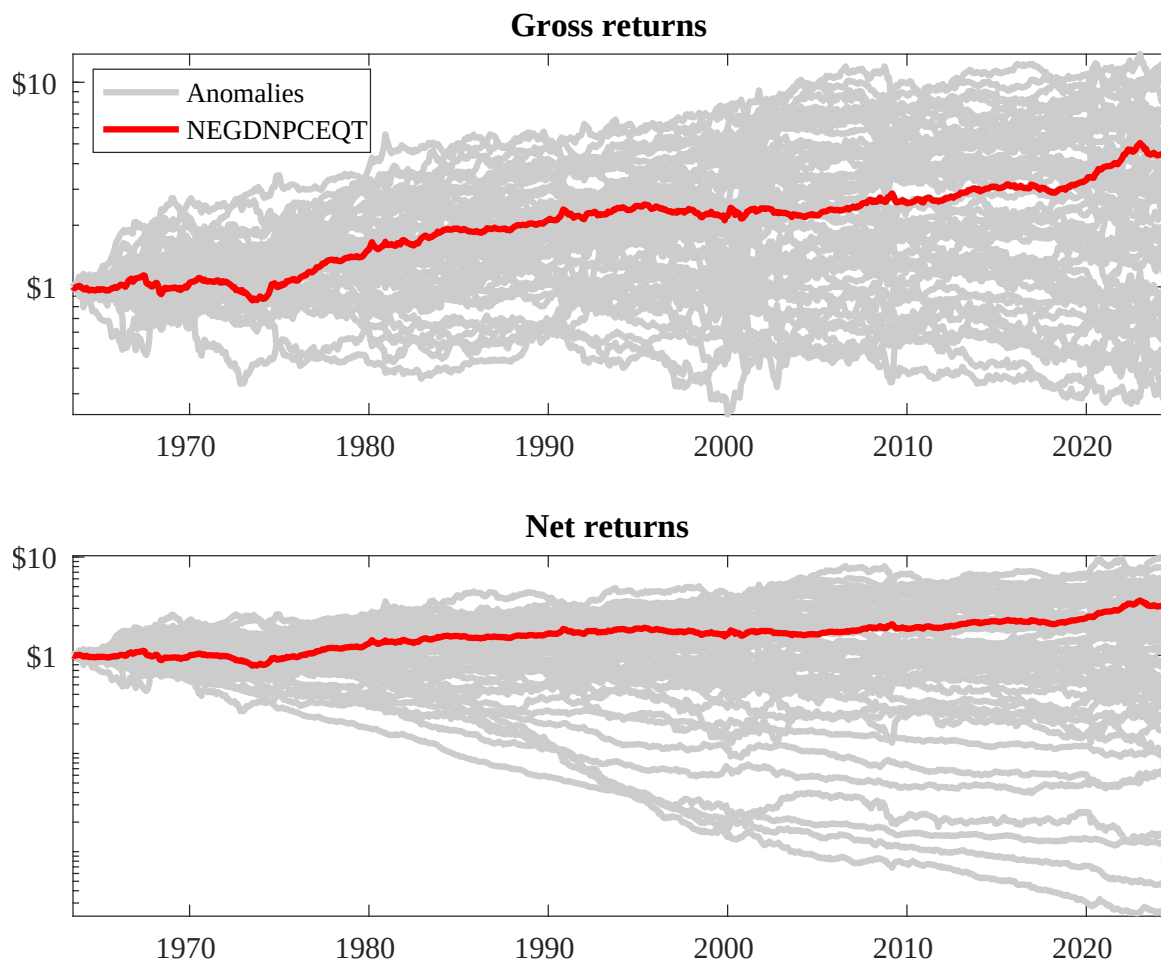
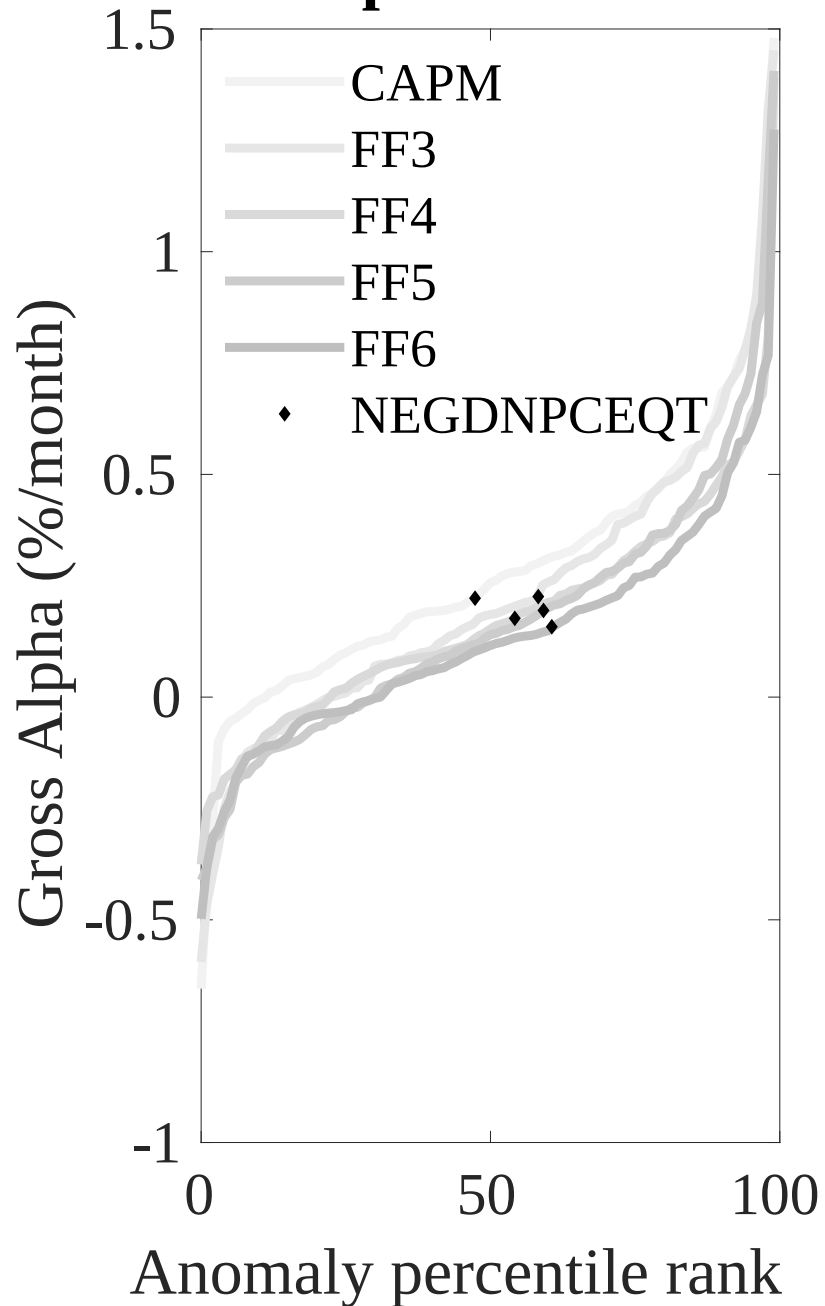


Figure 3: Dollar invested.

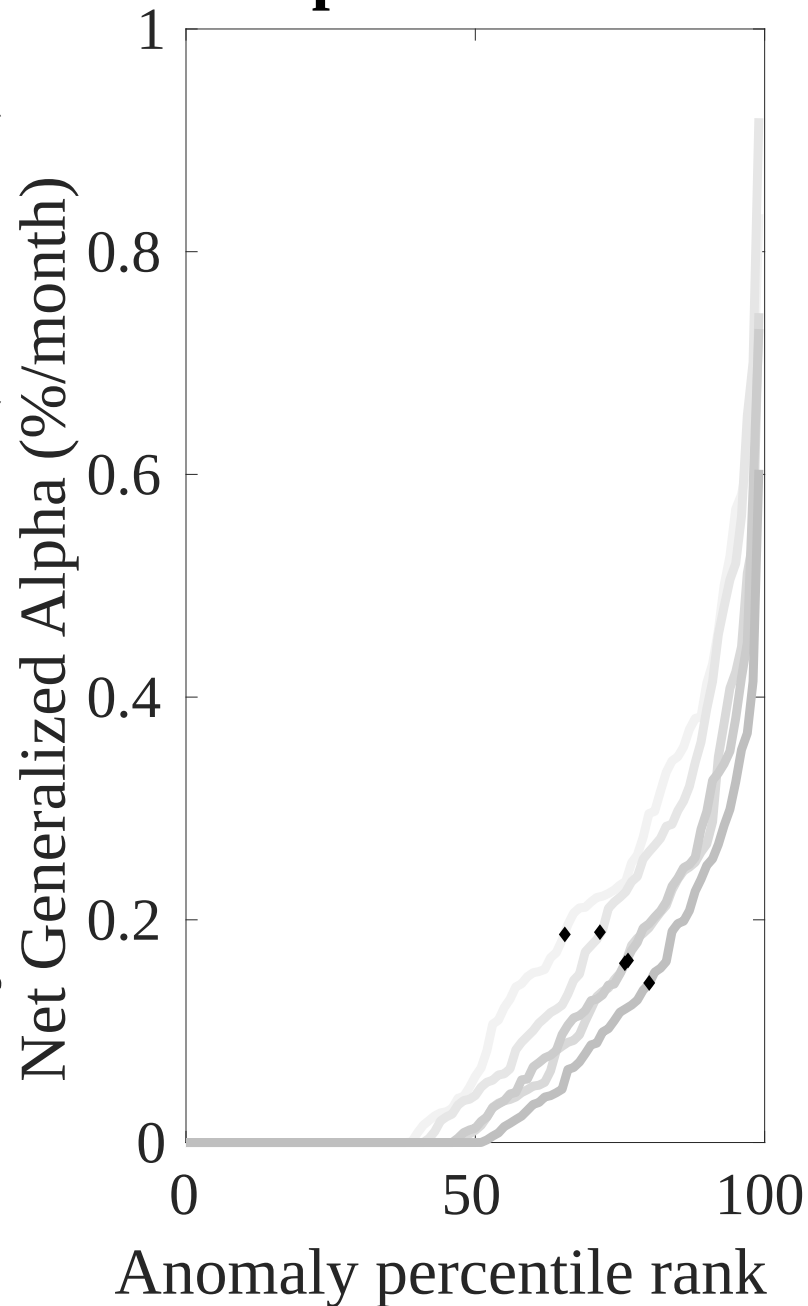
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the DCP trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

Gross Alpha Percentiles



Novy-Marx and Velikov (RFS, 2016)

Net Alpha Percentiles



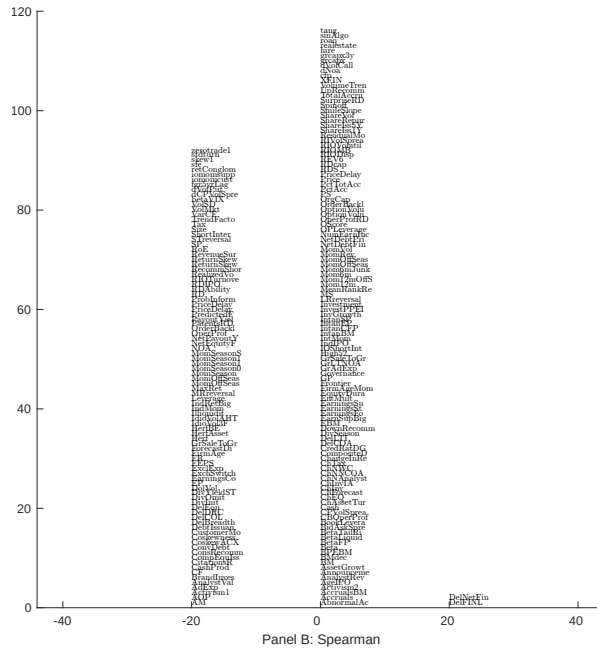
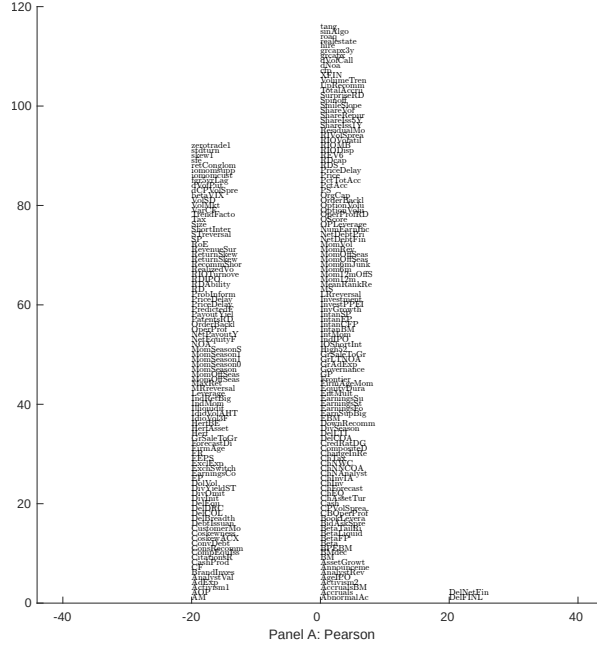


Figure 5: Distribution of correlations.

This figure plots a name histogram of correlations of 210 filtered anomaly signals with DCP. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

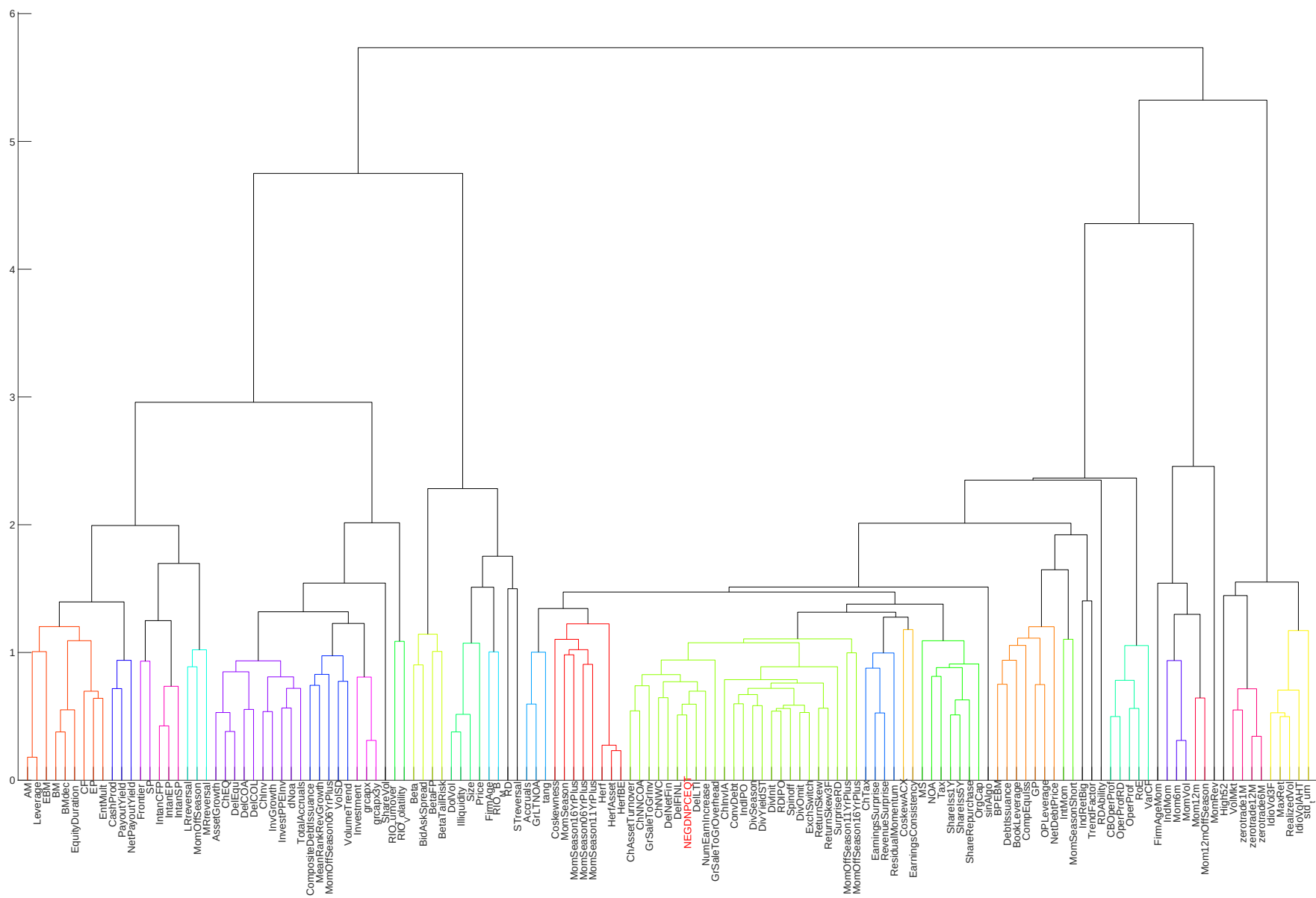


Figure 6: Agglomerative hierarchical cluster plot

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

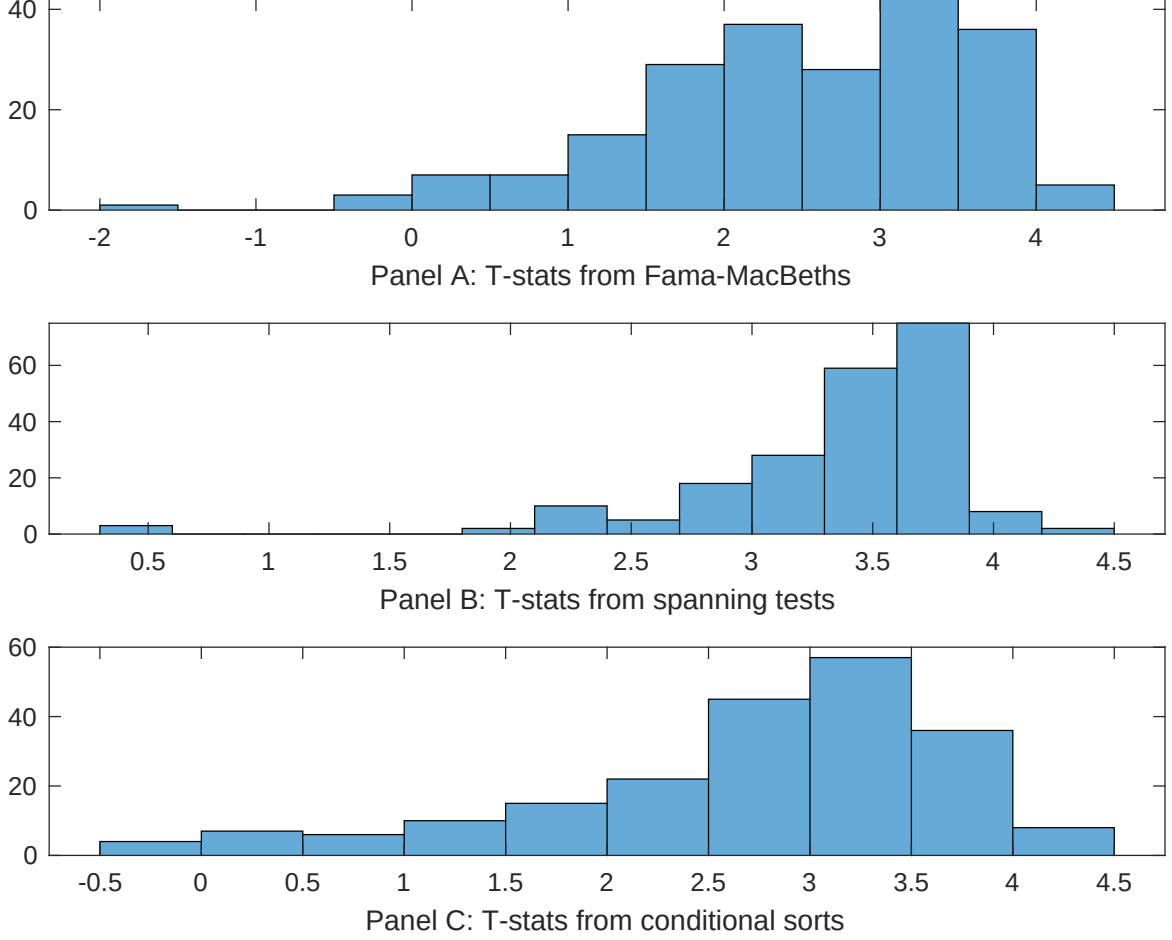


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of DCP conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{DCP} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{DCP} DCP_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{DCP,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on DCP. Stocks are finally grouped into five DCP portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted DCP trading strategies conditioned on each of the 210 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on DCP. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{DCP}DCP_{i,t} + \sum_{k=1}^s \beta_{X_k} X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Change in financial liabilities, Accruals, Change in Net Working Capital, Net debt financing, Investment to revenue, Change in Net Noncurrent Op Assets. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.12 [5.76]	0.11 [5.38]	0.12 [5.49]	0.12 [5.12]	0.15 [6.59]	0.12 [5.50]	0.14 [5.61]
DCP	0.69 [1.02]	0.15 [2.36]	0.20 [3.13]	0.14 [2.20]	0.13 [1.65]	0.22 [3.24]	-0.23 [-0.04]
Anomaly 1	0.17 [9.25]						0.59 [1.93]
Anomaly 2		0.13 [4.17]					0.21 [3.70]
Anomaly 3			0.62 [2.31]				-0.15 [-2.90]
Anomaly 4				0.21 [9.37]			0.90 [2.47]
Anomaly 5					0.24 [6.36]		0.15 [3.52]
Anomaly 6						0.75 [4.03]	0.16 [0.69]
# months	732	732	732	630	732	732	630
$\bar{R}^2(\%)$	0	0	0	0	0	0	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the DCP trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{DCP} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Change in financial liabilities, Accruals, Change in Net Working Capital, Net debt financing, Investment to revenue, Change in Net Noncurrent Op Assets. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.14 [2.42]	0.12 [2.02]	0.13 [2.24]	0.17 [2.79]	0.16 [2.63]	0.16 [2.77]	0.11 [2.00]
Anomaly 1	30.04 [9.09]						27.23 [6.18]
Anomaly 2		16.08 [6.80]					11.41 [4.61]
Anomaly 3			14.54 [4.92]				3.53 [1.12]
Anomaly 4				14.58 [4.28]			-8.49 [-2.04]
Anomaly 5					11.87 [4.95]		3.26 [1.39]
Anomaly 6						17.28 [6.46]	9.55 [3.37]
mkt	-1.72 [-1.27]	-1.20 [-0.86]	-2.60 [-1.84]	-1.01 [-0.70]	-2.63 [-1.87]	-2.47 [-1.77]	0.25 [0.18]
smb	4.67 [2.34]	11.02 [5.34]	8.33 [4.08]	7.25 [3.33]	5.39 [2.58]	6.90 [3.42]	7.46 [3.50]
hml	-8.32 [-3.15]	-7.40 [-2.71]	-9.60 [-3.51]	-9.80 [-3.56]	-9.81 [-3.60]	-11.30 [-4.19]	-5.71 [-2.19]
rmw	-7.50 [-2.82]	0.53 [0.19]	-3.43 [-1.24]	-5.61 [-1.98]	-3.95 [-1.43]	-7.00 [-2.56]	-2.31 [-0.83]
cma	10.59 [2.61]	17.35 [4.33]	21.33 [5.32]	18.18 [4.24]	20.80 [5.18]	21.94 [5.54]	10.40 [2.54]
umd	2.96 [2.15]	3.98 [2.86]	4.56 [3.24]	4.59 [3.18]	3.97 [2.79]	3.31 [2.33]	1.18 [0.86]
# months	732	732	732	630	732	732	630
$\bar{R}^2(\%)$	19	15	12	13	12	14	26

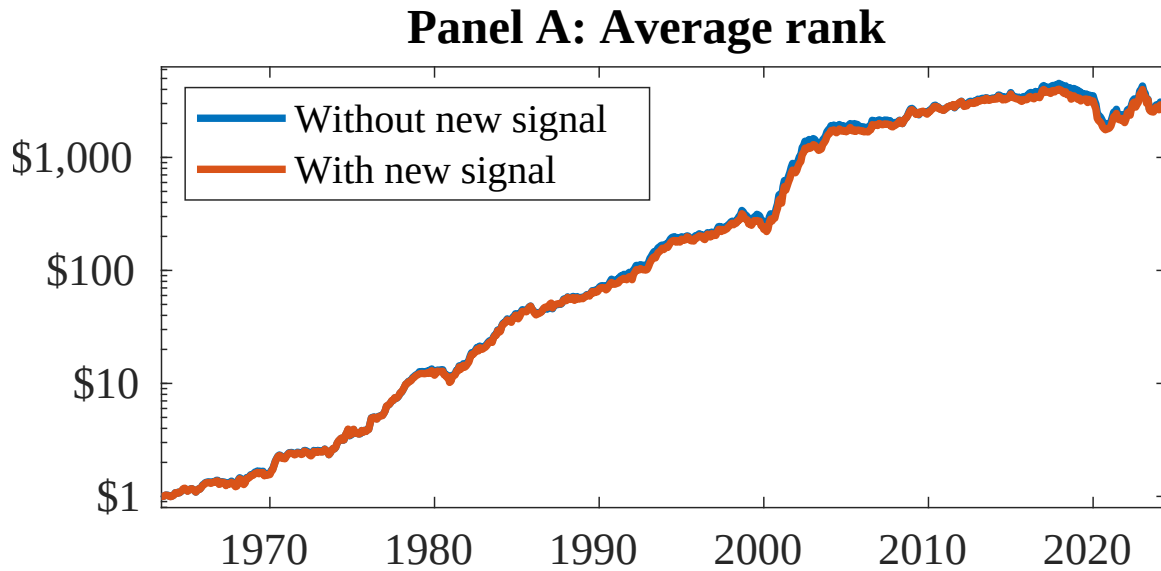


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 154 anomalies. The red solid lines indicate combination trading strategies that utilize the 154 anomalies as well as DCP. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.

References

- Bansal, R. and Yaron, A. (2004). Risks for the long run: A potential resolution of asset pricing puzzles. *Journal of Finance*, 59(4):1481–1509.
- Barro, R. J. (2006). Rare disasters and asset markets in the twentieth century. *Quarterly Journal of Economics*, 121(3):823–866.
- Breeden, D. T. and Litzenberger, R. H. (1978). Prices of state-contingent claims implicit in option prices. *Journal of Business*, 51(4):621–651.
- Campbell, J. Y. and Cochrane, J. H. (1999). By force of habit: A consumption-based explanation of aggregate stock market behavior. *Journal of Political Economy*, 107(2):205–251.
- Carhart, M. M. (1997). On persistence in mutual fund performance. *Journal of Finance*, 52:57–82.
- Chen, A. and Velikov, M. (2022). Zeroing in on the expected returns of anomalies. *Journal of Financial and Quantitative Analysis*, Forthcoming.
- Chen, A. Y. and Zimmermann, T. (2022). Open source cross-sectional asset pricing. *Critical Finance Review*, 27(2):207–264.
- Detzel, A., Novy-Marx, R., and Velikov, M. (2022). Model comparison with transaction costs. *Journal of Finance*, Forthcoming.
- Fama, E. F. and French, K. R. (1993). Common risk factors in the returns on stocks and bonds. *Journal of Financial Economics*, 33(1):3–56.
- Fama, E. F. and French, K. R. (2015). A five-factor asset pricing model. *Journal of Financial Economics*, 116(1):1–22.

- Fama, E. F. and French, K. R. (2018). Choosing factors. *Journal of Financial Economics*, 128(2):234–252.
- Jagannathan, R. and Wang, Y. (2007). Lazy investors, discretionary consumption, and the cross-section of stock returns. *Journal of Finance*, 62(4):1623–1661.
- Kroencke, T. A. and Julliard, C. (2012). Consumption-based asset pricing with higher cumulants. *Review of Economic Studies*, 79(4):1565–1598.
- Lettau, M. and Ludvigson, S. (2001). Consumption, aggregate wealth, and expected stock returns. *Journal of Finance*, 56(3):815–849.
- Livdan, D., Sapriza, H., and Zhang, L. (2009). Financially constrained stock returns. *Journal of Finance*, 64(4):1827–1862.
- Merton, R. C. (1974). On the pricing of corporate debt: The risk structure of interest rates. *Journal of Finance*, 29(2):449–470.
- Novy-Marx, R. and Velikov, M. (2016). A taxonomy of anomalies and their trading costs. *Review of Financial Studies*, 29(1):104–147.
- Novy-Marx, R. and Velikov, M. (2023). Assaying anomalies. *Working paper*.
- Parker, J. A. and Julliard, C. (2005). Consumption risk and the cross section of expected returns. *Journal of Political Economy*, 113(1):185–222.
- Rietz, T. A. (1988). The equity risk premium a solution. *Journal of Monetary Economics*, 22(1):117–131.